



JKBOSE 2026

CLASS 11TH

BINOMIAL THEOREM

MATHS

LECTURE-03

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TOPICS TO BE COVERED

1

Exercise Discussion 7.1 Part 2

2

Related Problems

QUESTION (NDA+JEE-MAINS 2018)

$${}^nC_r = {}^nC_{n-r}$$



What is the **ratio** of the **coefficient of x^{15}** to the **term independent of x** in $\left(x^2 + \frac{2}{x}\right)^{15}$?

1

$$\frac{1}{64} \times$$

2

$$\frac{1}{32} \checkmark$$

3

$$\frac{1}{16} \times$$

4

$$\frac{1}{4} \times$$

(Sol) Here $T_{r+1} = {}^{15}C_r (x^2)^{15-r} \left(\frac{2}{x}\right)^r$

$$\Rightarrow T_{r+1} = {}^{15}C_r 2^r x^{30-3r}$$

For x^{15} , we have
 $30 - 3r = 15$

$$\Rightarrow 3r = 15$$

$$\Rightarrow r = 5$$

So, Co-efficient of $x^{15} = {}^{15}C_5 (2)^5$

For Term independent of x ,

$$30 - 3r = 0$$

$$\Rightarrow r = 10$$

So, Term independent of $x = {}^{15}C_{10} (2)^{10}$

Therefore required Ratio.

$$= \frac{{}^{15}C_5 2^5}{{}^{15}C_{10} 2^{10}} = \frac{1}{2^5} = \frac{1}{32}$$

QUESTION (IIT-JEE 1999+JEE-MAINS 2019)

If in the expansion of $(1+x)^m(1-x)^n$ the coefficients of x and x^2 are 3 and -6 respectively, then the value of m is

1 ☒ 6

2 ☒ 9

3 ☒ 12

4 ☒ 24

(Sol) Using Binomial Theorem;

$$(1+x)^m(1-x)^n = \left(1+mx + \frac{m(m-1)}{2}x^2 + \dots\right) \left(1-nx + \frac{n(n-1)}{2}x^2 - \dots\right)$$

$$\Rightarrow (1+x)^m(1-x)^n = 1 + (m-n)x + \left(\frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn\right)x^2 + \dots$$

So, $m-n=3$ — (1)

$$\frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn = -6$$

$$\Rightarrow m^2 - m + n^2 - n - 2mn = -12$$

$$\Rightarrow m^2 - m + n^2 - n - 2mn = -12.$$

$$\Rightarrow m^2 + n^2 - 2mn - (m+n) = -12.$$

$$\Rightarrow (m-n)^2 - (m+n) = -12.$$

$$\Rightarrow 3^2 - (m+n) = -12 \text{ (By using ①)}$$

$$\Rightarrow -(m+n) = -12 - 9$$

$$\Rightarrow -(m+n) = -21$$

$$\Rightarrow m+n = 21 \text{ --- ②}$$

Adding ① & ②, we get;

$$m - \cancel{n} + m + \cancel{n} = 3 + 21$$

$$\Rightarrow 2m = 24$$

$$\Rightarrow m = 12, n = 9$$

EXAMPLE

Find the number of rational terms in the expansion of $(9^{\frac{1}{4}} + 8^{\frac{1}{6}})^{1000}$

(Sol) Here $T_{r+1} = {}^{1000}C_r (9^{\frac{1}{4}})^{1000-r} (8^{\frac{1}{6}})^r$

$$\Rightarrow T_{r+1} = {}^{1000}C_r (9)^{\frac{1000-r}{4}} (8)^{\frac{r}{6}}$$

$$\Rightarrow T_{r+1} = {}^{1000}C_r (3^{\frac{1}{2}})^{\frac{1000-r}{2}} (2^{\frac{1}{3}})^{\frac{r}{2}}$$

$$\Rightarrow T_{r+1} = {}^{1000}C_r (3)^{\frac{1000-r}{2}} (2)^{\frac{r}{2}}$$

The above term is rational only if the exponents of 3 & 2 are integers i.e. $\frac{1000-r}{2}$ & $\frac{r}{2}$ are integers. So the set of possible values of r is $\{0, 2, 4, 6, \dots, 1000\}$

Therefore the number of rational terms in the expansion of $(9^{\frac{1}{4}} + 8^{\frac{1}{6}})^{1000} = 501$.

Hence the result.

QUESTION(JEE-MAINS 2024)

Number of integral terms in the expansion of $(7^{\frac{1}{2}} + 11^{\frac{1}{6}})^{824}$ is equal to _____.

(Sol) Here, $T_{r+1} = {}^{824}C_r (7^{\frac{1}{2}})^{824-r} (11^{\frac{1}{6}})^r$

$\Rightarrow T_{r+1} = {}^{824}C_r (7)^{\frac{824-r}{2}} (11)^{\frac{r}{6}}$

The above term is integer only if the exponents of 7 and 11 are integers i.e. $\frac{824-r}{2}$ & $\frac{r}{6}$ are Integers. So the set of possible values of r is $\{0, 6, 12, 18, \dots, 822\}$

Therefore the number of Integral Terms in the expansion of $(7^{\frac{1}{2}} + 11^{\frac{1}{6}})^{824} = 138$.

- ✓✓ * Middle Terms
- ✓✓ * NGT (NUMERICAL Greatest Terms)

EXERCISE – 7.1

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$(1 - 2x)^5$$

(Dys)

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$\left(\frac{2}{x} - \frac{x}{2}\right)^5 \text{ (Dys)}$$

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$(2x - 3)^6$$

(Dys)

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$\left(\frac{x}{3} + \frac{1}{x}\right)^5$$

(DY5)

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$\left(x + \frac{1}{x}\right)^6$$

(Dys)

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$(96)^3$$
$$= (100 - 4)^3$$

(Dys)

The Magic You are looking
for is in the work you
are avoiding.

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$(102)^5 = (100 + 2)^5$$

(Dys).

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$(101)^4$$

$$= (100 + 1)^4$$

(Dys)

QUESTION

Expand each of the expressions in Exercises 1 to 5.

$$(99)^5 = (100-1)^5$$

Dys.

QUESTION

(Already done
through example).



Using Binomial Theorem, indicate which number is larger $(1.1)^{10000}$ or 1000.

$$(1+0.1)^{10000} >$$

QUESTION



Find $(a+b)^4 - (a-b)^4$. Hence, evaluate $(\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4$.

(Sol) From Binomial Theorem, we have;

$$(a+b)^4 = {}^4C_0 a^4 + {}^4C_1 a^3 b + {}^4C_2 a^2 b^2 + {}^4C_3 a b^3 + {}^4C_4 b^4.$$

$$\Rightarrow (a+b)^4 = a^4 + 4a^3 b + 6a^2 b^2 + 4ab^3 + b^4 \quad \text{--- (1)}$$

$$\text{Also } (a-b)^4 = a^4 - 4a^3 b + 6a^2 b^2 - 4ab^3 + b^4 \quad \text{--- (2)}$$

Subtracting (1) & (2), we get;

$$(a+b)^4 - (a-b)^4 = 8a^3 b + 8ab^3$$

$$\Rightarrow (a+b)^4 - (a-b)^4 = 8ab(a^2 + b^2)$$

$$\text{Hence } (\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4$$

$$= 8\sqrt{3} \times \sqrt{2} (\cancel{\sqrt{3}^2} + \cancel{\sqrt{2}^2})$$

$$= 40\sqrt{6}.$$

Hence the result.

QUESTION



Find $(x + 1)^6 + (x - 1)^6$. Hence or otherwise evaluate $(\sqrt{2} + 1)^6 + (\sqrt{2} - \sqrt{1})^6$.

h/w

QUESTION



Show that $9^{n+1} - 8n - 9$ is divisible by 64, whenever n is a positive integer.

Proof: we know that;

$$(a+b)^{n+1} = {}^{n+1}C_0 a^{n+1} + {}^{n+1}C_1 a^n b + {}^{n+1}C_2 a^{n-1} b^2 + \dots + {}^{n+1}C_{n+1} b^{n+1}.$$

Put $a=1, b=8$, we get

$$(1+8)^{n+1} = {}^{n+1}C_0 + {}^{n+1}C_1(8) + {}^{n+1}C_2(8)^2 + {}^{n+1}C_3(8)^3 + \dots + {}^{n+1}C_{n+1}(8)^{n+1}.$$

$$\Rightarrow 9^{n+1} = 1 + 8(n+1) + 8^2 \left\{ {}^{n+1}C_2 + {}^{n+1}C_3(8) + \dots + {}^{n+1}C_{n+1}8^{n-1} \right\}$$

$$\Rightarrow 9^{n+1} = 1 + 8n + 8 + 64q$$

$$\Rightarrow 9^{n+1} - 8n - 9 = 64q$$

$\Rightarrow 9^{n+1} - 8n - 9$ is divisible by 64, where $n \in \mathbb{N}$.

Hence proved.

QUESTION

Prove that $\sum_{r=0}^n 3^r nC_r = 4^n$.

Proof: From Binomial Theorem, we have;

$$(a+b)^n = \sum_{r=0}^n nC_r a^{n-r} b^r$$

Put $a=1, b=3$, we get

$$(1+3)^n = \sum_{r=0}^n nC_r (1)^{n-r} (3)^r$$

$$\sum_{r=0}^n 3^r nC_r = 4^n$$

Hence proved.

Thank
You